1(a)	$f(-1) = (-1)^4 + a(-1)^3 - 3(-1)^2 + b(-1) + 5 = 4$	M1
	$1 - a - 3 - b + 5 = 4 \Rightarrow a + b = -1$ *	A1*
		(2)
(b)	$f(2) = (2)^4 + a(2)^3 - 3(2)^2 + b(2) + 5 = -23$	M1
	$\Rightarrow 8a + 2b = -32$ oe (eg $4a + b = -16$)	A1
	$b = -1 - a \Rightarrow 4a - 1 - a = -16 \Rightarrow a = ...$	dM1
	$a = -5, b = 4$	A1
		(4)
		(6 marks)

2(a)	$\left(\frac{dy}{dx} =\right) \ 3x^2 - 2x - 16$	M1A1
	$3x^2 - 2x - 16 = 0 \Rightarrow x = ...$	M1
	$x = \frac{8}{3}, -2$	A1
		(4)
(b)	$\left(\frac{d^2y}{dx^2} =\right) \ Cx + D$	M1
	$\Rightarrow \frac{d^2y}{dx^2} = C \times "..." + D(>0) \Rightarrow \min \text{ or } (<0) \Rightarrow \max$	dM1
	All three components: $\left(\frac{d^2y}{dx^2} =\right) \ 6x - 2$ when $x = \frac{8}{3}, \ \frac{d^2y}{dx^2} = ...(=14) > 0 \Rightarrow \min$ when $x = -2, \ \frac{d^2y}{dx^2} = ...(=-14) < 0 \Rightarrow \max$	A1
		(3)
Alt (b)	E.g. $x = -2.5, \frac{dy}{dx} = \frac{31}{4}, x = 0, \frac{dy}{dx} = -16$	M1
	E.g. $x = -2.5, \frac{dy}{dx} = \frac{31}{4} > 0, x = 0, \frac{dy}{dx} = -16 < 0$ so $x = -2$ is a maximum $\left(x = 0, \frac{dy}{dx} = -16 < 0\right), x = 3, \frac{dy}{dx} = 5 > 0,$ so $x = \frac{8}{3}$ is a minimum	dM1 A1
		(3)
		(7 marks)

3(i)	$7^{x+2} = 3 \Rightarrow 7^2 \times 7^x = 3$	M1
	$\Rightarrow x = \log_7 \frac{3}{7^2}$	A1
	$x = \log_7 \frac{3}{49}$	A1
		(3)
Alt(i)	$7^{x+2} = 3$	
	$(x+2)\log_7 7 = \log_7 3$	M1
	$\Rightarrow x = \log_7 3 - 2$	A1
	$x = \log_7 \frac{3}{49}$	A1
		(3)
(ii)	$1 + \log_2 y + \log_2 (y+4) = \log_2 (5-y)$	
	E.g. $1 + \log_2 y(y+4) = \log_2 (5-y)$ or $1 \rightarrow \log_2 2$	M1
	$\log_2 \left(\frac{y(y+4)}{5-y} \right) = -1 \Rightarrow \frac{y(y+4)}{5-y} = \frac{1}{2}$ or $\log_2 \left(2y(y+4) \right) = \log_2 (5-y) \Rightarrow 2y(y+4) = 5-y$	dM1
	$2y^2 + 9y - 5 = 0$	A1
	$(2y-1)(y+5) = 0 \Rightarrow y = ...$	ddM1
	$y = \frac{1}{2}$	A1
		(5)
		(8 marks)

4(a)	$(2+px)^6 = (2^6+)\ 6\times 2^5(px)+\frac{6\times 5}{2}\times 2^4(px)^2+\ldots$ or $\left(1+"\frac{p}{2}"x\right)^6 = 1+6\times\left("\frac{p}{2}"x\right)+\frac{6\times 5}{2}\times\left("\frac{p}{2}"x\right)^2+\ldots$	M1
	2^6 or 64	B1
	$+192px$ or $+240p^2x^2$	A1
	$(64+)192px+240p^2x^2$	A1
		(4)
(b)	$\left(3-\frac{1}{2}x\right)(2+px)^6 \Rightarrow \left(3-\frac{1}{2}x\right)(64+192px+240p^2x^2)$	
	Attempts $3\times"240p^2"$ and $\left(-\frac{1}{2}\right)\times"192p"$	M1
	$3\times"240p^2"+\left(-\frac{1}{2}\right)\times"192p"=-\frac{3}{4}$	dM1
	$2880p^2-384p+3=0 \Rightarrow p=\ldots$	ddM1
	$(p=)\ \frac{1}{8},\frac{1}{120}$	A1
		(4)
		(8 marks)

5(i)	(As $x \geqslant 0$ so $\sqrt{3x}$ exists and) $\left(\sqrt{3x}-1\right)^2 \geqslant 0$	M1
	Hence $3x-2\sqrt{3x}+1 \geqslant 0$	M1
	$\Rightarrow 3x+1 \geqslant 2\sqrt{3x}$ *	A1*
		(3)
Alt 1	$3x+1 \geqslant 2\sqrt{3x} \Leftrightarrow (3x+1)^2 \geqslant 12x \Leftrightarrow 9x^2-6x+1 \geqslant 0$	M1
	$9x^2-6x+1 \geqslant 0 \Leftrightarrow (3x-1)^2 \geqslant 0$	M1
	Square numbers are greater than or equal to zero so $(3x-1)^2 \geq 0$ is true hence $3x+1 \geqslant 2\sqrt{3x}$ *	A1*
		(3)
Alt 2	If $3x+1 < 2\sqrt{3x}$ then $3x-2\sqrt{3x}+1 < 0$	M1
	So $(...\sqrt{3x} \pm ...)^2 < 0$ or $(\sqrt{x} \pm ...)^2 < 0$	M1
	But $(\sqrt{3x}-1)^2 \geq 0$ for all $x \geqslant 0$ so $3x+1 \geqslant 2\sqrt{3x}$	A1
(ii)	Shows that it is not true for three consecutive prime numbers Eg $5+7+11=23$ which is not divisible by 5 (so not true)	B1
		(1)
		(4 marks)

6(a)	$\frac{3\sin\theta\cos\theta}{2\sin\theta-1}=5\tan\theta \Rightarrow \frac{3\sin\theta\cos\theta}{2\sin\theta-1}=5\underline{\frac{\sin\theta}{\cos\theta}}$	M1
	$\frac{3\sin\theta\cos\theta}{2\sin\theta-1}=5\frac{\sin\theta}{\cos\theta} \Rightarrow 3\sin\theta\cos^2\theta=5\sin\theta(2\sin\theta-1)$	M1
	$3\sin\theta\cos^2\theta=10\sin^2\theta-5\sin\theta \Rightarrow 3\sin\theta(1-\sin^2\theta)=10\sin^2\theta-5\sin\theta$ *	M1
	$\Rightarrow 3\sin^3\theta+10\sin^2\theta-8\sin\theta=0$	A1*
		(4)
(b)	$3\sin^3 2x+10\sin^2 2x-8\sin 2x=0 \Rightarrow \sin 2x(3\sin^2 2x+10\sin 2x-8)=0$ $(3\sin 2x-2)(\sin 2x+4)=0 \Rightarrow \sin 2x=...$	M1
	$\sin 2x=\frac{2}{3}$	A1
	$(x=)0.365$	A1
	$x=0$	B1
		(4)
		(8 marks)

7(a)	$(t = 10, P =)$ 1.2	B1
	$(t = 11.5, P =)$ 2.821	B1
		(2)
(b)	$h = 0.5$	B1
	$E \approx \left(\frac{1}{2} \times "\frac{1}{2}"\right)\left["1.2" + 2.95 + 2(1.882 + 2.45 + "2.821")\right]$	M1 A1ft
	$E \approx 4.61$ (kWh)	A1
		(4)
		(6 marks)

8(a)	$a_2 = 2p^2 - 7$	B1
		(1)
(b)	$a_3 = 2("2p^2 - 7"+3)^2 - 7$	M1
	$p-3+2p^2-7+8p^4-32p^2+25 = p+15$ $\Rightarrow 8p^4 - 30p^2 = 0$	M1 A1
	Correct values of $p^2 = 0$ and $\frac{15}{4}$ (may be implied by $p = 0$ and $\pm\sqrt{\frac{15}{4}}$)	A1
	Uses their values of p to find a_2 using a correct method	dM1
	Possible values for a_2 are -7 and $\frac{1}{2}$	A1
		(6)
Alt (b)	$\sum_{n=1}^{3} a_n = p+15 \Rightarrow p-3+a_2+2(a_2+3)^2-7 = p+15$	M1
	$\Rightarrow a_2 + 2(a_2+3)^2 - 25 = 0$ $\Rightarrow 2a_2{}^2 + 13a_2 - 7 = 0$	M1A1 A1
	$\Rightarrow (2a_2-1)(a_2+7) = 0 \Rightarrow a_2 = ...$	dM1
	Possible values for a_2 are -7 and $\frac{1}{2}$	A1
		(6)
		(7 marks)

9(a)(i)	Centre = $(k, 2k)$	B1
(ii)	Radius = $\sqrt{k+7}$	B1
		(2)
(b)(i)	$(2-k)^2+(3-2k)^2=k+7 \Rightarrow 4-4k+k^2+9-12k+4k^2=k+7$	M1
	$5k^2-17k+6=0$ *	A1*
(ii)	$(k=)\frac{2}{5}, 3$	B1
		(3)
	Centre is $\left(\frac{2}{5}, \frac{4}{5}\right)$	B1ft
	Gradient of tangent $\pm\frac{2-"\frac{2}{5}"}{3-"\frac{4}{5}"} = \left(-\frac{8}{11}\right)$	M1
	$y-3="-\frac{8}{11}"(x-2) \Rightarrow$ sets $y=0 \Rightarrow x=...$ Alternatively, $\tan\angle PTO=\frac{8}{11} \Rightarrow XT=\frac{3}{8/11}=...$ where X is (2,0)	M1
	Area $OPT=\frac{1}{2}\times"\frac{49}{8}"\times 3=...$ Alternatively, Area $OPT=\frac{1}{2}\times 2\times 3+\frac{1}{2}\times"\frac{33}{8}"\times 3=...$	dM1
	$=\frac{147}{16}$ oe	A1
		(5)
		(10 marks)

10(a)	$d = \frac{37-15}{11} (= 2)$	M1
	$u_5 = 15 + 4 \times "2" = ...$	M1
	Week 5 Sunday run = 23 (km)	A1
		(3)
(b)	$r^{11} = \frac{37}{15} \Rightarrow r = \sqrt[11]{\frac{37}{15}} = ...$	M1
	$u_5 = 15 \times "1.0855..."^4 = ...$	M1
	Week 5 Sunday run = awrt 20.8 (km)	A1
		(3)
(c)	$S_{12} = \frac{12}{2}[2 \times 15 + (12-1) \times "2"]$ or $\frac{12}{2}(15+37)$ $(= 312)$	M1
	$S_{12} = \frac{15(1-"1.08554.."^{12})}{1-"1.08554.."}$ $(= 294.185..)$	M1
	Difference = $\left(\frac{360-312}{12}\right)$ or $\left(\frac{360-294.185...}{12}\right) = ...$	dM1
	Either Training model A: $x \leqslant$ 4 km, Training model B: $x \leqslant$ 5.4… km with = or $\leqslant$	A1
	Training model A: x = 4 km, Training model B: x = 5km	A1cso
		(5)
		(11 marks)